

CV

PROFILE

20+ years building production systems across Switzerland, Germany, Portugal and Brazil. Specialised in software engineering, data visualization, cloud infra and AI/LLM integrations.

Fluent English • German B1 TELC • EU citizen • Swiss B-Visa • Available

TECH SKILLS

Git • Database modeling and SQL • SaaS • Data Science and Visualization • Web Development • UX • Rapid Prototyping • AI Agents • System analysis and modeling with UML • MVC system architecture • Web crawlers and bots for Data Mining and ETL • APIs • Linux • VPS • SSH • DevOps • CI/CD • Javascript Frameworks • Ruby • Python • Containerization • OWASP

PROFESSIONAL EXPERIENCE

04/2025 – present

Freelance Data & Visualisation Engineer • Basel, CH

- Concepts for large-scale cartography platforms
- Mapping of territories, networks, and flows
- Innovative system for offline navigation

02/2024 – 03/2025

DevOps Engineer • Swisscom • Zürich, CH

- Development of new system features with Ruby on Rails
- Deployment with Docker and Kubernetes

02/2023 – 02/2024

Senior Tech Lead • EDGE Strategy • Zug, CH

- Hiring and managing developers
- Design and development of new systems, e.g., TicketSystem, Offline PayTool and EDGEv2 Platform
- Infrastructure upgrades and cybersecurity

01/2008 – 01/2023

Freelance IT Engineer • ExtraPolo • various countries

- Requirements analysis with clients
- 40+ web projects developed



Rafael Polo

Address

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Date of Birth

13.06.1985

Nationality

Portuguese and Brazilian
Permit B

Website

<https://extrapolo.com/>

02/2014 – 06/2016

Full-Stack Developer • Tactical Tech Collective • Berlin, DE

- Data visualization tools for investigative journalists
- Hands-on workshop on security and data exploration

11/2013 – 02/2014

Software Engineer • Trommsdorf + Drüner • Berlin, DE

- Python and Ruby systems programming
- Database optimization and modelling

02/2007 – 08/2008

Software Engineer • Cortex Intelligence • Rio de Janeiro, BR

- Database modeling for dashboards development
- Development of crawlers for ETL

FORMAL EDUCATION

2010 – 2012

MSc Information Systems

Universidade Federal do Estado do Rio de Janeiro, BR

Research Area: Knowledge Representation and Reasoning

2003 – 2009

BSc Computer Science

Universidade Metodista Bennett, Rio de Janeiro, BR

Thesis: Agile Systems Development for Web 2.0

LANGUAGES

German	B1 TELC certificate
English	Fluent
Spanish	Fluent
Greek	A2 certified
Portuguese	Native

About me

I am a curious, creative, and collaborative engineer who thrives at the intersection of data, software, and science - building tools that turn complexity into insight.

Volunteer Work

I helped create a Computer Lab focused on young refugees, including a special Kids@Labs day in Athens (2017–2018), and taught programming basics.

Website

Further original projects, web-selected works, exhibitions, and awards available at my interactive website →



<https://extrapolo.com/>

RELEVANT PROJECTS

SwissViz - 2026

Technologies: TypeScript, D3.js, deck.gl, MapLibre

All Swiss companies as single interactive map, adapted to mobile and filtered by cantons and economic activities. Updated daily.

<https://rafapolo.github.io/swissviz/>

VISO - 2026

Technologies: TypeScript, D3.js, WASM, WebGL

Interactive Visualization of Obscure Systems - is an Interactive and framework for visualizing and analyzing parliamentary expenses, also allowing users to run SQL queries offline.

<https://rafapolo.github.io/viso/>

Ξ.xyz - 2025

Technologies: TypeScript, D3.js, Altair, Plotly

My hub of published data visualizations, enabling exploration of entire cities, economic networks, and demographic distributions.

<https://Ξ.xyz/dataviz>

FinCrime - 2025

Technologies: TypeScript, D3.js, WebGL

In partnership with Brazilian journalists, this platform allows the fast visualization of companies networks among related fields: economic areas, owners, partners or addresses.

<https://rafapolo.github.io/fincrimel/>
